

# Cost Allocation via the Nucleolus

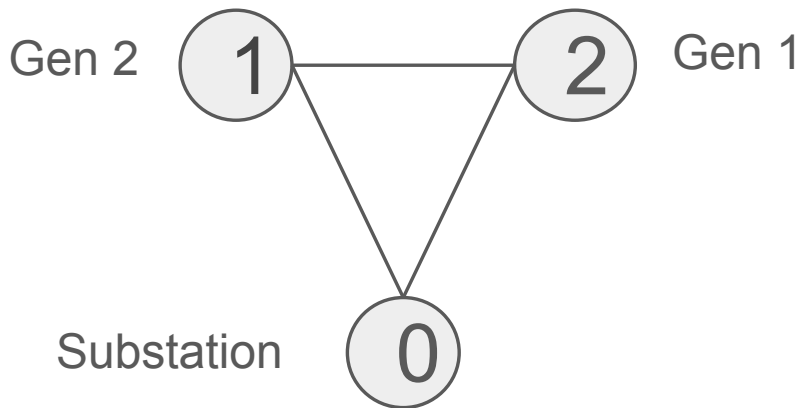
Project 2: Econ 238

# Cost Allocation of Connecting Renewables to Grid

## Objective Function

$$\min \sum_{e \in \mathcal{E}} \text{INV}(e) \cdot \text{cap}(e) + G \cdot P$$

where  $P = 0$ , so the  $G \cdot P$  term vanishes



1. Power Flow Constraints at each generation node  $i \in N$  and substation node

$$\sum_{l \in \delta^+(i)} f_{lt} - \sum_{l \in \delta^-(i)} f_{lt} = s_i \cdot g_{i,t} \quad \forall i \in S, \forall t \in T$$

$$\sum_{l \in \delta^-(0)} f_{lt} - \sum_{l \in \delta^+(0)} f_{lt} = \sum_{i \in S} s_i \cdot g_{i,t} \quad \forall t \in T$$

2. Line Capacity Limits for each  $l \in L$  and  $t \in T$

$$-F_l \leq f_{lt} \leq F_l \quad \forall l \in L, \forall t \in T$$

3. Peak injection lower bound, for each  $t \in T$

$$G \geq \sum_{i \in S} s_i \cdot g_{i,t} \quad \forall t \in T$$

4. Non-negativity and free flow

$$F_l \geq 0 \quad \forall l \in L, \quad f_{lt} \in \mathbb{R} \quad \forall l, t, \quad G \geq 0.$$

# Cost Allocation via the Nucleolus - Methodology

## Coalition Costs $C(s)$

- The cost for the coalition, subset  $S$ , to operate independently

## $\sum x_i$

- The total share of the "grand coalition" cost

$$\sum x_i \leq C(S) \text{ for all } S \subsetneq N$$

## Objective LP

- Find  $C(s)$  for every coalition

## Core

- no sub-coalition has an incentive to break away because they can't achieve a lower cost on their own
- If cost allocation is inside core, no subgroup can find cheaper alternative.
- There are a spectrum of allocations

# Cost Allocation via the Nucleolus - Methodology

## Excess - Maximize the Minimum

$$e(S, \mathbf{x}) = C(S) - \sum_{i \in S} x_i$$

## Subject To Constraints

1. excess of **every** possible sub-coalition is greater than or equal to this floor:  
 **$e(s, \mathbf{x}) \geq \epsilon$**

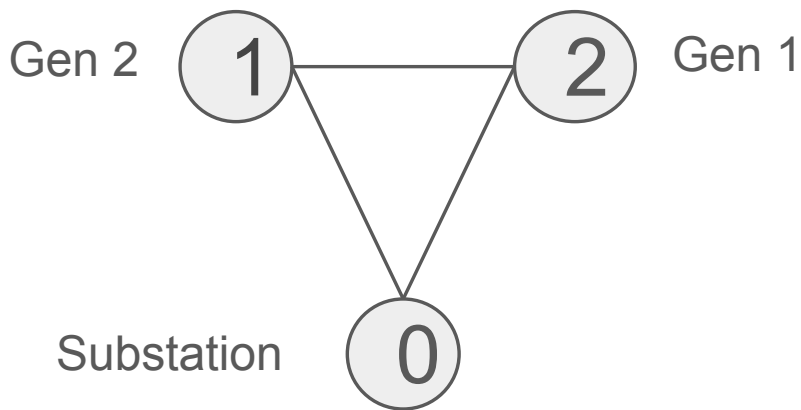
## NUCLEOLUS

1. Define excess  $e(S, \mathbf{x})$ 
  - a. How much coalition  $S$  saves compared to acting alone.
2. Maximize the minimum  $\epsilon$ 
  - a. Finding the allocations  $\mathbf{x}$  that maximizes  $\epsilon$  (min excess) such that every coalition saves at least  $\epsilon^*$
  - b. Coalitions hitting  $\epsilon^*$  are frozen for next LP
3. Lexicographic Refinement
  - a. Binding coalitions are frozen and among remaining allocations we maximize the next floor until they hit the next  $\epsilon^*$  hit
4. Iterated until all generators are fixed.

# Problem 1

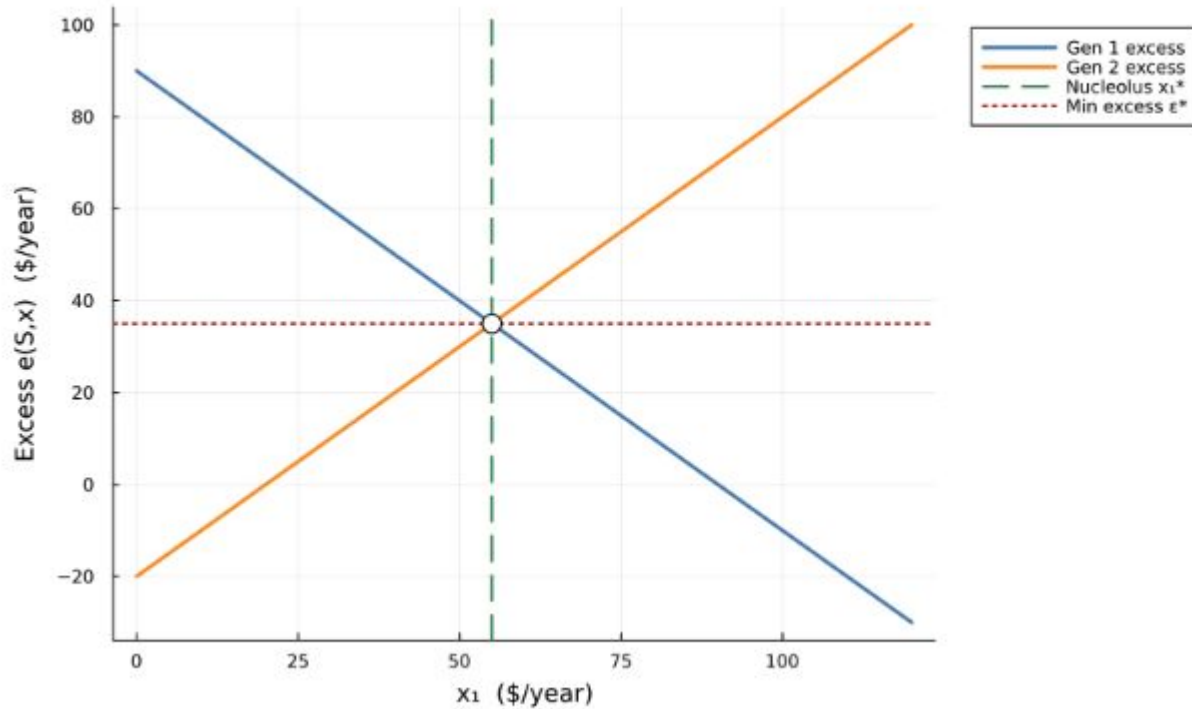
**Perfect anti-correlation:  $n = 2$ ,  $T = 2$**

## Perfect anti-correlation: $n = 2$ , $T = 2$



| Item             | Value         |
|------------------|---------------|
| Time periods (T) | 2             |
| Generators (N)   | 2             |
| $g_1$ matrix     | [0, 1]        |
| $g_2$ matrix     | [1, 0]        |
| INV(1, 0)        | \$90 / MW·yr  |
| INV(2, 0)        | \$100 / MW·yr |
| INV(1, 2)        | \$50 / MW·yr  |

# Results and Discussion



| Coalition      | Cost (\$/yr) |
|----------------|--------------|
| C({1})         | \$90.00      |
| C({2})         | \$100.00     |
| C({1,2})       | \$120.00     |
| Standalone sum | \$190.00     |
| Total savings  | \$70.00      |

## Results and Discussion

Discussion: The two generators face standalone transmission costs of \$90/yr and \$100/yr respectively, totalling \$190/yr. However, by cooperating, they reduce this to \$120/yr, which saves them a combined total of \$70/yr. Gen 1 pays \$55/yr and Gen 2 pays \$65/yr. Hence, Gen 1 saves \$35/yr and Gen 2 also saves \$35/yr, so neither player loses.

| Coalition      | Cost (\$/yr) |
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| $C(\{2\})$     | \$100.00     |
| $C(\{1,2\})$   | \$120.00     |
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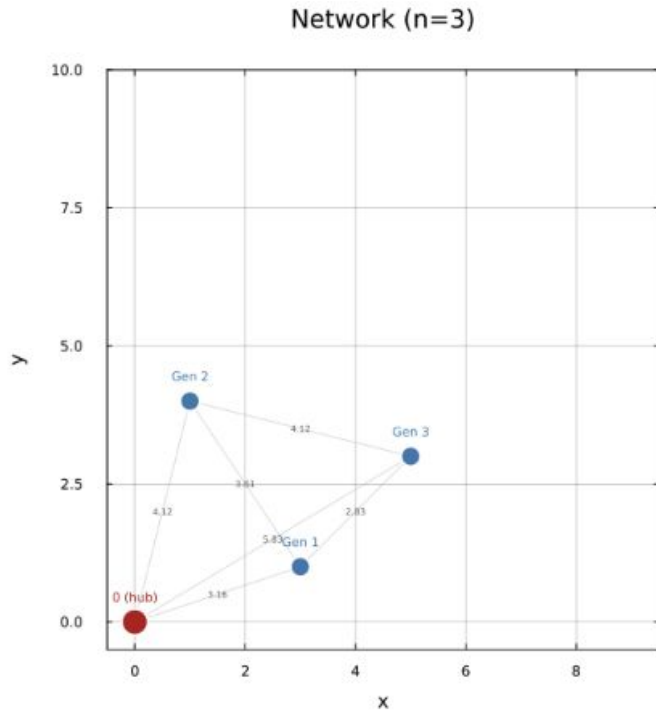
| Generator | $x^*$ (\$/yr) | Share of $C(N)$ | Saves   | Reduction |
|-----------|---------------|-----------------|---------|-----------|
| Gen 1     | \$55.00       | 45.8%           | \$35.00 | 38.9%     |
| Gen 2     | \$65.00       | 54.2%           | \$35.00 | 35.0%     |
| Sum       | \$120.00      | = $C(N)$        | \$70.00 |           |



# Problem 2

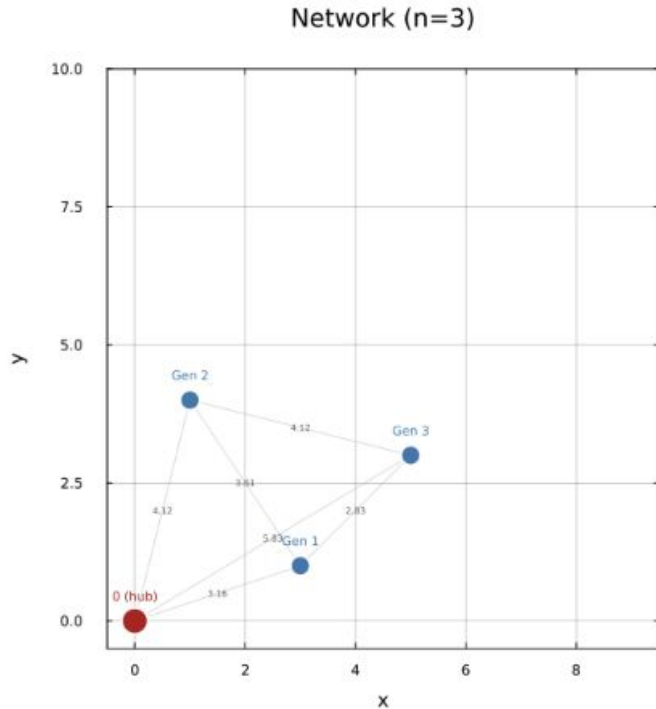
**Scaling:  $n = 3$ , and  $n = 10$ ,  $T = n$**

## Scaling: $n = 3$ , and $n = 10$ , $T = n$



|                             | Value                                        |
|-----------------------------|----------------------------------------------|
| Time periods (T)            | 3                                            |
| Generators (N)              | 3                                            |
| $g_1$                       | [1, 0, 0]                                    |
| $g_2$                       | [0, 1, 0]                                    |
| $g_3$                       | [0, 0, 1]                                    |
| Investment cost $INV(i, j)$ | Euclidean distance between nodes $i$ and $j$ |
| Generation cost (P)         | 0                                            |

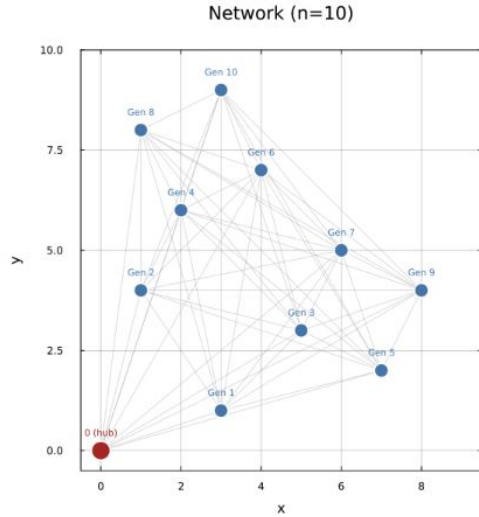
## Scaling: $n = 3$ , and $n = 10$ , $T = n$



| Generator | Allocation $x^*$ | Standalone | Saves | Saves % |
|-----------|------------------|------------|-------|---------|
| Gen 1     | 1.47             | 3.16       | 1.70  | 53.6%   |
| Gen 2     | 2.60             | 4.12       | 1.53  | 37.1%   |
| Gen 3     | 3.06             | 5.83       | 2.78  | 47.5%   |
| Total     | 7.12             | 13.12      | 5.99  | 45.7%   |

|                       | Value  |
|-----------------------|--------|
| Standalone cost       | 13.12  |
| Coalition cost $C(N)$ | 7.12   |
| Total savings         | 5.9979 |
| Savings %             | 45.7%  |

# Scaling: $n = 3$ , and $n = 10$ , $T = n$



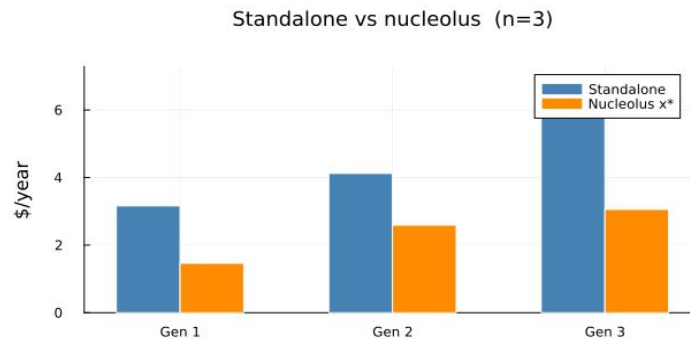
|                       | Value   |
|-----------------------|---------|
| Standalone cost       | 68.6869 |
| Coalition cost $C(N)$ | 14.2974 |
| Total savings         | 54.3895 |
| Savings %             | 79.1%   |

| Generator | Allocation $x^*$ | Standalone | Saves   | Saves % |
|-----------|------------------|------------|---------|---------|
| Gen 1     | 1.1608           | 3.1623     | 2.0015  | 63.3%   |
| Gen 2     | 1.0982           | 4.1231     | 3.0249  | 73.4%   |
| Gen 3     | 1.5516           | 5.8310     | 4.2793  | 73.4%   |
| Gen 4     | 1.3170           | 6.3246     | 5.0076  | 79.2%   |
| Gen 5     | 1.7358           | 7.2801     | 5.5443  | 76.2%   |
| Gen 6     | 1.1132           | 8.0623     | 6.9491  | 86.2%   |
| Gen 7     | 1.1132           | 7.8102     | 6.6971  | 85.7%   |
| Gen 8     | 1.7358           | 8.0623     | 6.3264  | 78.5%   |
| Gen 9     | 1.7358           | 8.9443     | 7.2084  | 80.6%   |
| Gen 10    | 1.7358           | 9.4868     | 7.7510  | 81.7%   |
| Total     | 14.2974          | 68.6869    | 54.3895 | 79.1%   |

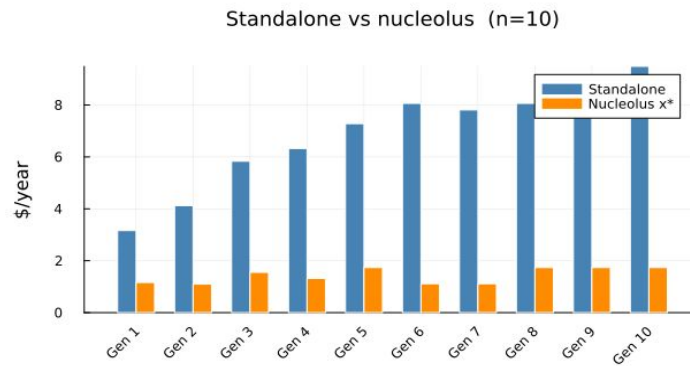
# Discussion

- In both cases, every single generator pays less under the nucleolus than going alone.
- The savings grow dramatically with  $n$  because larger coalitions unlock more network sharing opportunities.

$n = 3$



$n = 10$



# Problem 3

**Correlation Study:  $n=2$ ,  $T=168$  hours**

## Statistical Framework & Methodology (n=2, T=168 h)



### Bivariate Uniform (Copula)

Controlled via Gaussian Copula to prescribe correlation while maintaining **0–100 MW** marginal capacity bounds for both generators.



### Mersenne Twister PRNG

High-quality pseudorandom generator (MT19937) chosen for its massive period, ensuring statistical independence across 168 hours. Chose 42 as the random seed number (standard dummy seed in CS).

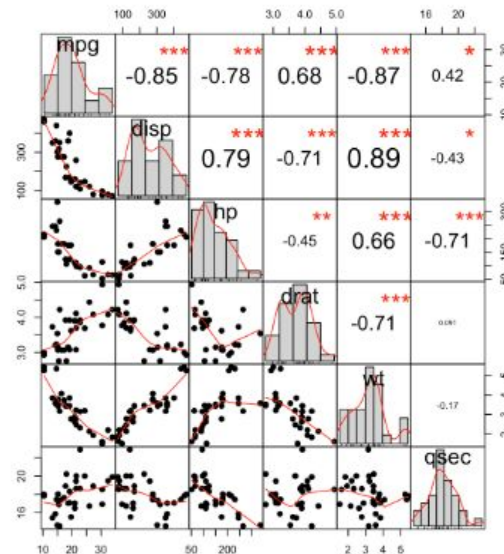
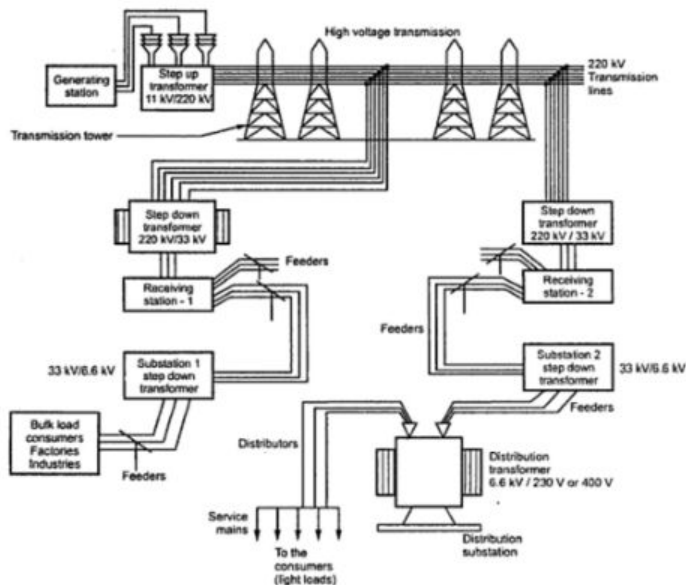


Figure 1: Sample Bivariate Intensity Mapping

# Correlation Analysis & Allocation



## Economic Key Takeaways

**Monotonicity of Savings:** Cooperative savings decrease as  $p$  to 1\$. When profiles align, peak capacity requirements cannot be offset, nullifying joint gains.

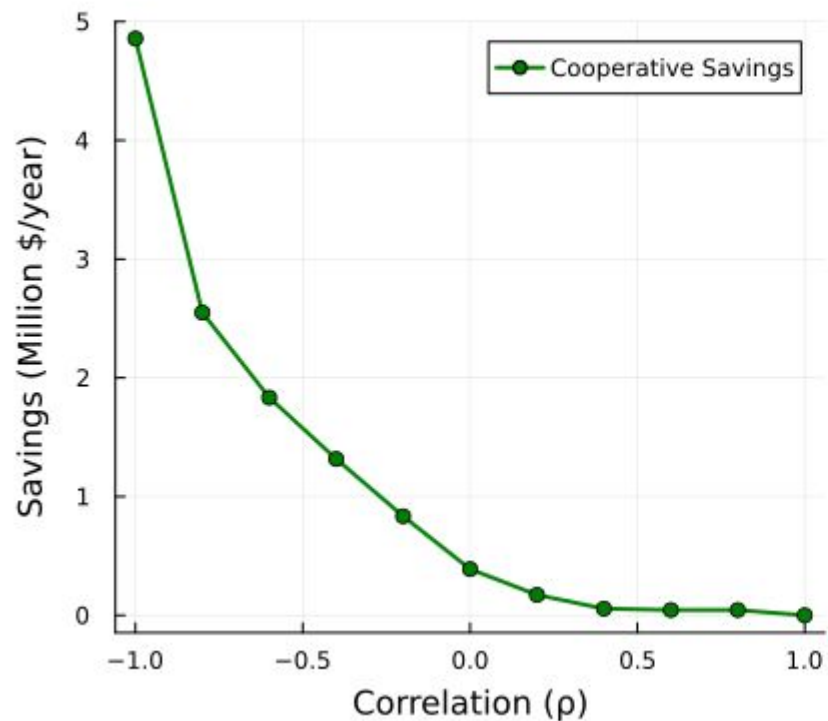
**The Nucleolus Signal:** Because generators are symmetric (same marginals), the nucleolus allocates cost shares at **50/50**, splits savings precisely, and remains stable across  $p$ .

**Gains & Losses:** Both generators gain significantly at  $p = -1$  (anti-correlated). At  $p = 1$ , the "coalition" is merely two identical lines, and cooperative value is lost.

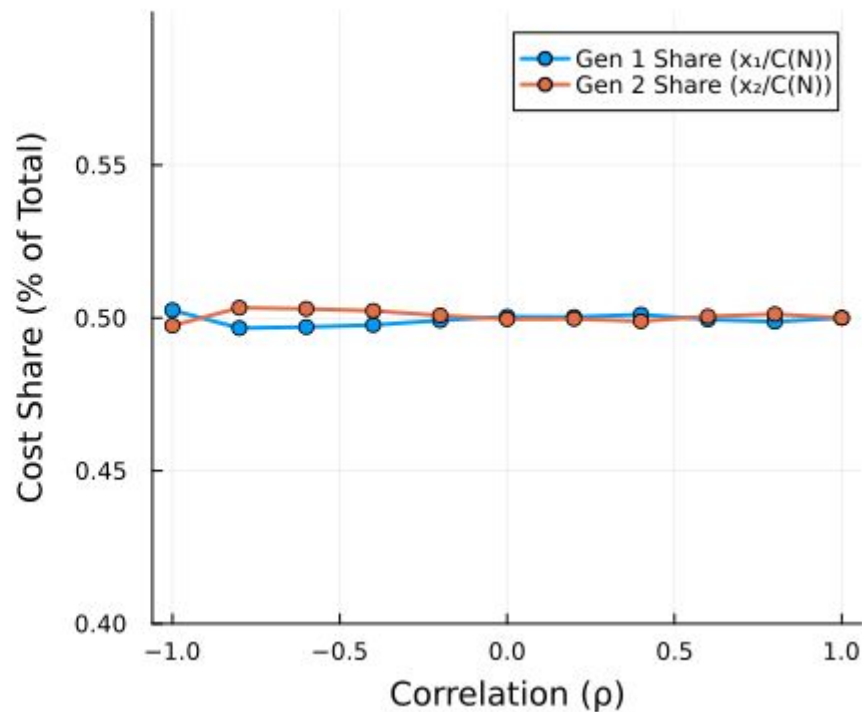
| Parameter              | Value / Range            | Economic Significance                                      |
|------------------------|--------------------------|------------------------------------------------------------|
| Correlation ( $\rho$ ) | -1.0 to +1.0             | Determines the magnitude of peak-shaving potential.        |
| Savings Curve          | Monotone Decr.           | Reflects the diminishing returns of shared infrastructure. |
| Nucleolus $x^*$        | $\sim \$0.5 \times C(N)$ | Truthful cost signal for symmetric, shared-use assets.     |



### Savings vs Correlation



### Nucleolus Allocation vs Correlation



| $\rho$ | C({1}) (\$)  | C({2}) (\$)  | C({1,2}) (\$) | Savings (\$) | $x1*/C(N)$ | $x2*/C(N)$ |
|--------|--------------|--------------|---------------|--------------|------------|------------|
| -1.0   | 4,941,414.34 | 4,916,059.63 | 5,000,000.00  | 4,857,473.98 | 0.5025     | 0.4975     |
| -0.8   | 4,941,414.34 | 4,990,484.66 | 7,382,450.95  | 2,549,448.05 | 0.4967     | 0.5033     |
| -0.6   | 4,941,414.34 | 4,989,629.66 | 8,097,710.81  | 1,833,333.19 | 0.4970     | 0.5030     |
| -0.4   | 4,941,414.34 | 4,981,231.27 | 8,606,069.66  | 1,316,575.95 | 0.4977     | 0.5023     |
| -0.2   | 4,941,414.34 | 4,956,047.21 | 9,063,028.32  | 834,433.24   | 0.4992     | 0.5008     |
| 0.0    | 4,941,414.34 | 4,933,345.32 | 9,485,604.87  | 389,154.79   | 0.5004     | 0.4996     |
| 0.2    | 4,941,414.34 | 4,935,339.47 | 9,705,152.60  | 171,601.21   | 0.5003     | 0.4997     |
| 0.4    | 4,941,414.34 | 4,920,033.84 | 9,805,450.37  | 55,997.82    | 0.5011     | 0.4989     |
| 0.6    | 4,941,414.34 | 4,951,361.88 | 9,848,259.89  | 44,516.33    | 0.4995     | 0.5005     |
| 0.8    | 4,941,414.34 | 4,965,928.25 | 9,862,826.26  | 44,516.33    | 0.4988     | 0.5012     |
| 1.0    | 4,941,414.34 | 4,941,414.34 | 9,882,828.69  | 0.00         | 0.5000     | 0.5000     |

# Discussion

- **Monotonic decay (for cooperation):** As the correlation coefficient ( $\rho$ ) increases toward 1.0, cooperative savings decrease monotonically, highlighting the economic necessity of combining diverse or anti-correlated generation profiles.
- **Stability of the nucleolus allocation:** The symmetric capacity limits (100 MW) ensure that the nucleolus allocates cost shares equally (a 50/50 split) across all correlation levels. This provides a highly stable and robust cost-allocation signal regardless of the underlying correlation.
- **Peak-Shaving infrastructure efficiency:** The total cost of the grand coalition  $C(\{1,2\})$  is highly sensitive to the coincidence of peak output. Anti-correlated outputs reduce the requirement for reinforced, high-capacity transmission lines.
- **Financial implications:** At  $\rho=1.0$ , the savings drop to zero because the joint line capacity equals the sum of the individual capacities. This reveals the threshold where the coalition loses its financial advantage over standalone development.
- **Grid planning strategies:** There is large financial value in resource diversity. Integrating uncorrelated or anti-correlated renewable resources reduces the stress on the transmission network and lowers total capacity expansion costs.

# Problem 4

## **Binary Investment Decisions**

# Binary Investment Model Set-Up

## Changes to Objective Function and Constraints

$$\min \sum_{e \in \mathcal{E}} y_e \cdot \text{INV}_e \cdot K_e^{\text{fixed}} + P \cdot G$$

subject to:

$$f_{e,t} \leq y_e \cdot K_e^{\text{fixed}} \quad \forall e \in \mathcal{E}, t \in \mathcal{T} \quad (1)$$

$$f_{e,t} \geq -y_e \cdot K_e^{\text{fixed}} \quad \forall e \in \mathcal{E}, t \in \mathcal{T} \quad (2)$$

$$y_e \in \{0, 1\} \quad \forall e \in \mathcal{E} \quad (3)$$

## Consequences from binary variable introduction:

- MILP! Convexity is lost!
- If  $\epsilon^* > 0$ , cooperation is still stable.

## Expected changes to model outputs

- “Lumpiness” in results → the continuous model builds exactly as much capacity as the worst case flow requires. The binary model must build a whole unit or nothing.
- Result → overbuilding and less savings from cooperation.

# Test Example: $N = 2$ , $K_{\text{fixed}} = 1.0$ , $g_{1,2}=1$ , $g_{2,1}=1$

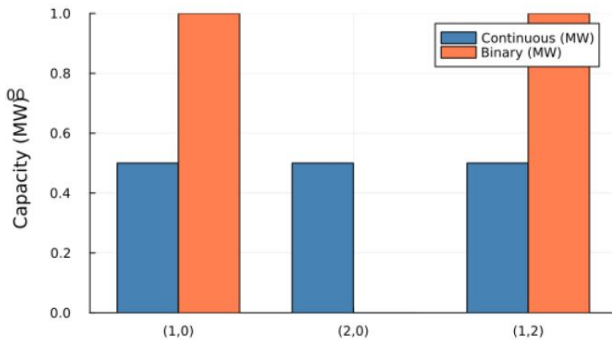
## Continuous Model

- When cooperating, 3 lines built:
  - (1,0) = 0.5 MW
  - (2,0) = 0.5 MW
  - (1,2) = 0.5 MW

## Binary Model

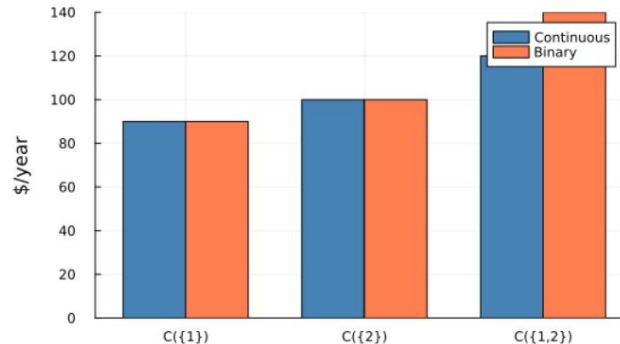
- When cooperating, 2 lines built:
  - (1,0) = 1 MW
  - (2,0) = 0 MW (not built)
  - (1,2) = 1 MW

Line Capacities for Grand Coalition



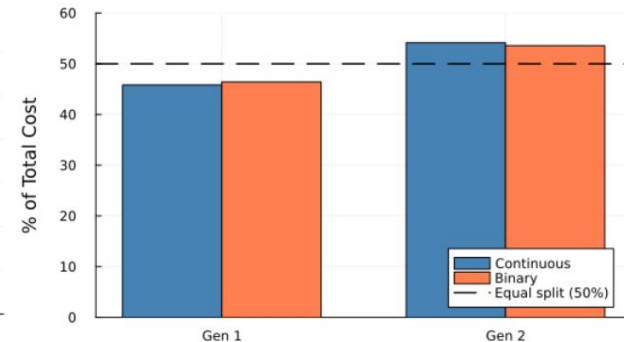
Line capacities change!

Coalition Costs: Continuous vs Binary



Coalition costs go up!

Cost Allocation Shares



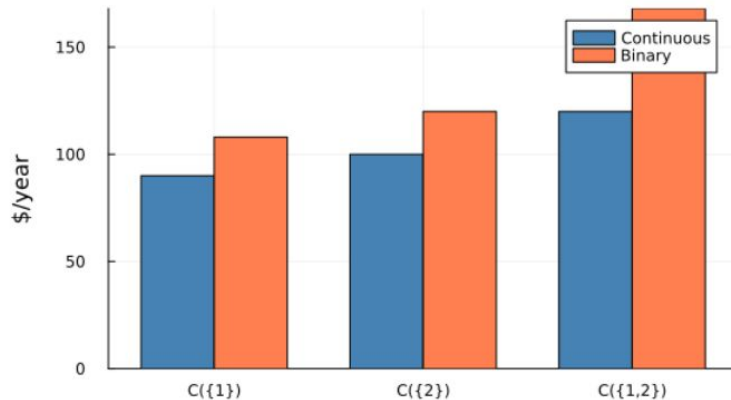
Cost allocations change slightly.

# Expanded Test: $N = 2$ , $K_{\text{fixed}} = 0.9$ vs $K_{\text{fixed}} = 1.2$

## $K_{\text{fixed}} = 1.2$

- Continuous: (0.5, 0.5, 0.5)
- Binary: (1.2, 0, 1.2)
- Cost savings: \$60

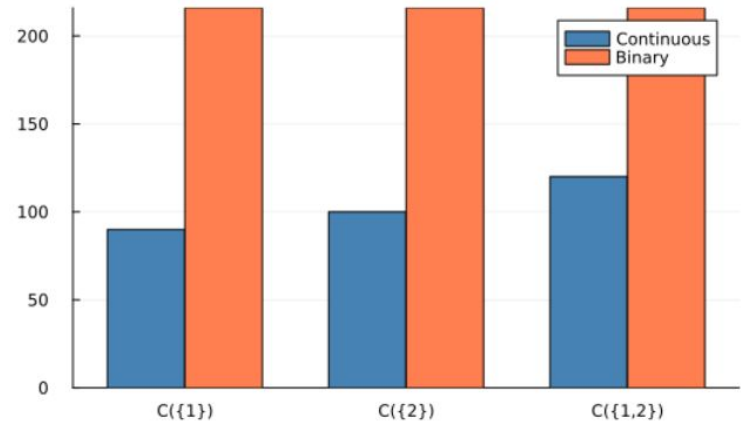
Coalition Costs: Continuous vs Binary



## $K_{\text{fixed}} = 0.9$

- Continuous: (0.9, 0.9, 0.9)
- Binary: (0.9, 0.9, 0.9)
- Cost savings: depends on how you want to consider it...

Coalition Costs: Continuous vs Binary

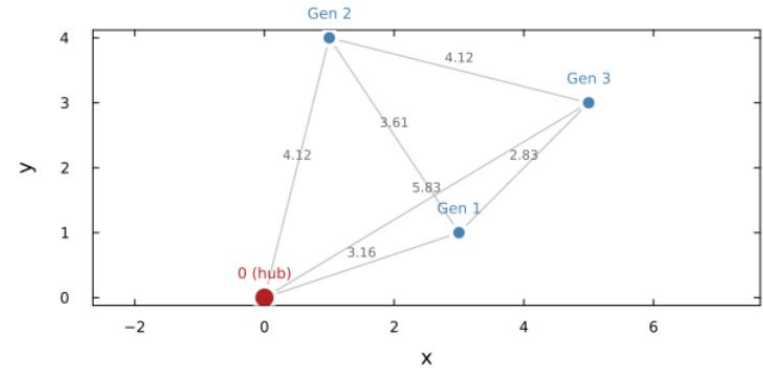


# Scaling Up Test: $N = 3$ , $K=1.0$

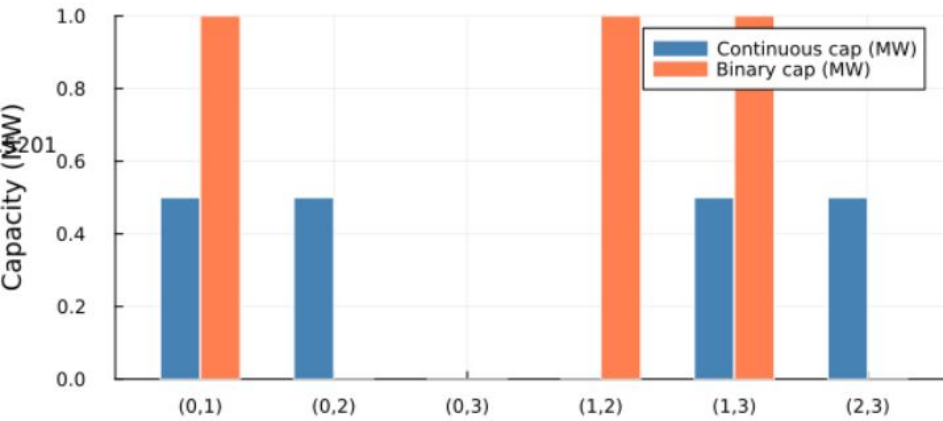
## Key Takeaways of Binary vs Continuous:

- 3 large lines built instead of 4 smaller lines;
- All coalition costs are larger under binary modeling than continuous modeling;
- Cost allocations shift slightly.

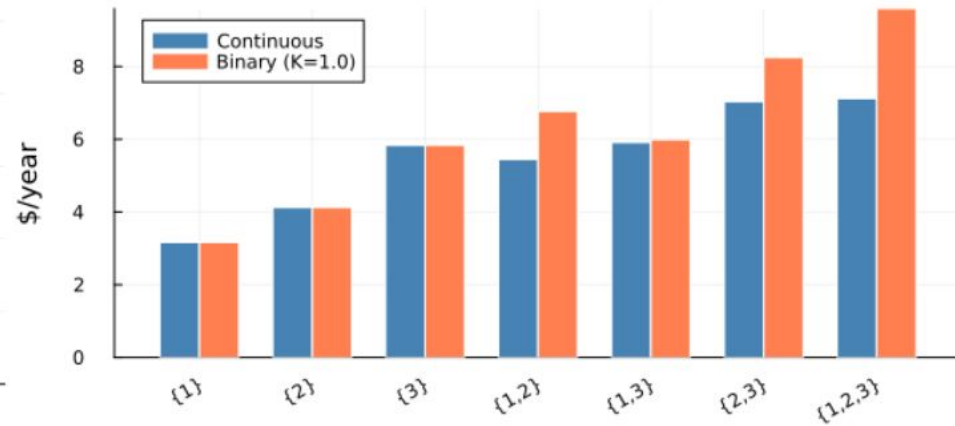
Network topology ( $n=3$ )



Grand Coalition {1,2,3}  
Line Capacities Built



Coalition Costs: Continuous vs Binary

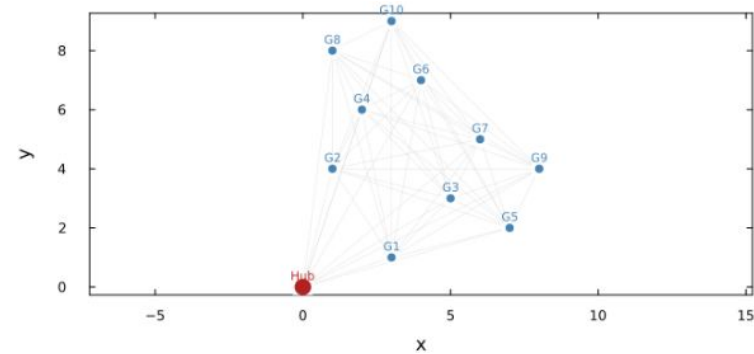




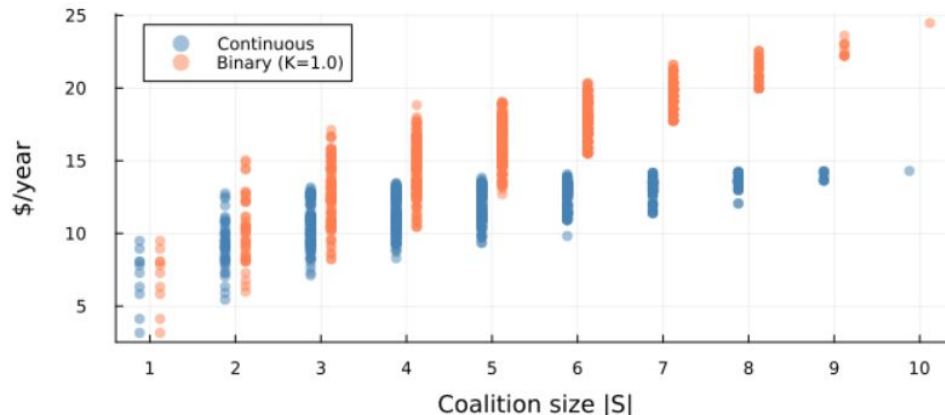
# Scaling Up: $N = 10$ , $K=1.0$

## Key Takeaways of Binary vs Continuous:

- 10 large lines built instead of 11 smaller lines;
- All coalition costs are larger under binary modeling than continuous modeling;
- Cost allocations shift slightly.



Coalition Costs by Size  
(Continuous vs Binary)



Nucleolus Allocation  
Continuous vs Binary

